When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation

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Abstract

In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, documents form interdependent networks — yet the relationships between them are represented as textual references rather than navigable structures. When these references are incomplete or invisible, the burden of reconstructing pathways falls on practitioners, who become the missing infrastructure between documents. This paper argues that traceability is not only a documentation property but a condition of interaction, and that navigation behavior is an empirical trace of the structural conditions practitioners are working through. I report two prototype-based studies conducted with $N = 20$ GMP-experienced participants. In the first, I modeled a synthetic GMP document corpus as a directed graph using NetworkX, identifying 31 missing CAPA links, 103 broken downstream links, and 21 lifecycle mismatch risks, and developed an external traceability visibility prototype that made linkage structure navigable without modifying validated source files. In the second, I developed a behavioral navigation tracking prototype that recorded document transitions and loopback events during a standardized out-of-specification investigation scenario. Findings show that traceability visibility reduced path reconstruction time by 30–60% and that loopback behavior — observed across all sessions — was attributable to document structure rather than interface design, with less experienced participants exhibiting more loopbacks and attributing navigation

difficulty to SOP structure rather than to the prototype. I introduce loopback tracking as an empirical method for surfacing document-structural friction, identify traceability as a first-class HCI design problem in regulated environments, and argue that making document relationships visible is a form of redistributing epistemic access — reducing the gap between practitioners who carry implicit navigational knowledge and those who do not.

Keywords: document traceability, loopback behavior, GMP, pharmaceutical manufacturing, navigation friction, epistemic access, regulated environments, invisible work

1 Introduction

In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, documents do not exist in isolation. A single deviation investigation may require navigating across standard operating procedures, analytical method protocols, validation records, and specification documents — each of which references others through textual cross-references rather than navigable links. When these references are complete and current, traceability is possible. When they are missing, outdated, or structurally ambiguous, the burden of reconstructing pathways falls entirely on the practitioner. The practitioner becomes, in effect, the missing infrastructure between documents.

This paper argues that traceability is not only a property of documentation — whether references exist, whether audit trails can be reconstructed — but a condition of interaction. When document relationships are only partially visible, practitioners must navigate through incomplete structures, and their navigation behavior reflects the structure they are working through. A loopback — a return to a previously visited document — is not simply evidence of inefficient searching. It is evidence that the path forward was not yet legible: that the document system did not resolve the judgment demand on the first pass.

I report a two-part study that connects document structure to navigation behavior in GMP environments. In the first part, I constructed a synthetic GMP document corpus and analyzed it as a directed graph using Python and NetworkX, identifying missing links, outdated references, and lifecycle mismatches. I then developed an external traceability

visibility prototype and evaluated it with $N = 20$ QA/QC-experienced participants on path completion tasks. In the second part, I developed a behavioral navigation tracking prototype that records document transitions, visit frequency, and loopback events during a standardized out-of-specification (OOS) investigation scenario, and evaluated it with $N = 20$ GMP-experienced participants. Neither prototype modifies validated source documents — a constraint constitutive of GMP-regulated environments.

Findings from the two studies establish a structural-behavioral relationship: document linkage failure is not only a maintenance problem but a navigation burden that propagates into observable behavior. Loopbacks, repeated transitions, and extended path reconstruction are not random inefficiencies — they are structured responses to unresolved information states produced by partially visible document networks. Making linkage structure visible reduces this burden; tracking navigation behavior makes it observable.

The contributions of this work are threefold. First, I introduce document traceability as an HCI problem specific to regulated environments, identifying the no-modification constraint as a novel design boundary for document navigation research. Second, I demonstrate that behavioral navigation tracking — loopback analysis in particular — can function as an empirical method for surfacing document-structural friction without deploying instruments inside validated systems. Third, I offer a theoretical framing that connects document architecture to navigation behavior to accountability conditions: in GMP environments, whether a practitioner can reconstruct a defensible pathway depends not only on their expertise but on whether the document system makes its own structure legible.

2 Related Work

Documents as Organizational Action

The foundational premise of this study — that document systems do not merely store information but actively organize the conditions under which work can be done — draws on Harper's (1998) ethnographic account of documentation practices at the IMF. Harper

shows that documents are not neutral containers: their accessibility, their linkage to other documents, and the navigational demands they impose on users are constitutive of organizational action. When document relationships are visible and traversable, practitioners can reconstruct accountable pathways; when they are not, that reconstruction work falls to individuals. Applied to GMP environments, this framework implies that traceability is not simply a property of document content but of document architecture — and that architectural failures redistribute labor in ways that are rarely measured.

Articulation Work and the Human as Missing Infrastructure

Schmidt & Bannon (1992) introduce the concept of articulation work to describe the coordinative labor that makes other work possible: the activities required to manage the distributed, interdependent nature of cooperative work that formal systems do not support. Articulation work is typically invisible — it is performed in the gaps between official procedures, not captured by them. In regulated document environments, cross-reference reconstruction is a form of articulation work: when linkage structure is incomplete or invisible, practitioners must manually bridge the gaps between documents, inferring relationships from partial cues and memory. This bridging labor is not represented in any quality record. The present study treats this invisible bridging as an object of design inquiry — asking whether making linkage structure visible can reduce the articulation work that practitioners currently absorb.

Information Foraging and the Cost of Navigation

Pirolli & Card (1999) develop information foraging theory to explain how people navigate information environments: users follow information scent — proximal cues about the value and accessibility of information sources — and move between information patches in ways that maximize gain relative to cost. A loopback, in this framework, is not an error but a rational response to weak or misleading scent: when the path forward does not resolve the task, returning to a previously visited document is the appropriate behavior. Applied to GMP document navigation, this implies that loopbacks are structural signals

rather than individual failures — they indicate that the document system did not provide sufficient scent to support forward navigation, and that the practitioner was forced to reorient from a prior position. The behavioral tracking study in this paper operationalizes this insight: loopback frequency is used as an empirical measure of navigation friction produced by incomplete linkage architecture.

Situated Action and the Conditions of Document Navigation

Suchman's (1987) account of situated action establishes that navigation through a document system is not plan-driven but shaped by the immediate conditions of execution — what is visible, accessible, and interpretable at the moment a decision must be made. A practitioner conducting an OOS investigation does not follow a pre-specified path through the document system; they navigate opportunistically, following available cues and adapting to what they find. When cross-references are broken, outdated, or ambiguous, the conditions of navigation are degraded, and the practitioner must substitute inferential work for structural guidance. This study examines the behavioral traces of that substitution — the loopbacks, repeated transitions, and extended path reconstructions that emerge when document architecture fails to support the situated action it was designed to enable.

Invisible Work and the Unmeasured Labor of Path Reconstruction

Star & Strauss (1999) show that work performed in the gaps of formal systems tends to become invisible: it is not captured by indicators, not attributed to the structural conditions that produce it, and not recognized as labor by the organizations that depend on it. In regulated document environments, the manual reconstruction of broken or incomplete traceability pathways is precisely this kind of invisible work. A practitioner who spends time reconstructing a cross-reference chain that should have been navigable has performed real labor — but that labor is not recorded anywhere in the quality system. The present study makes this labor visible in two ways: through graph analysis of linkage structure, which identifies where invisible bridging is likely to be required, and through behavioral tracking, which records the navigation patterns that invisible bridging produces.

Prototype-Based Evaluation

The methodological approach builds on established practice for prototype-based evaluation in HCI. Nielsen (1994) demonstrates that think-aloud evaluation with small participant samples reliably surfaces usability and navigability problems. Hutchinson et al. (2003) establish the logic of using a prototype simultaneously as a design intervention and as an analytic probe — a technology probe makes the conditions of use visible rather than simply testing interface quality. The present study applies this dual function to regulated document navigation: the traceability visibility prototype and the behavioral tracking prototype are evaluated not only as interfaces but as instruments for making document-structural friction visible and measurable without deployment inside validated systems.

Prior work has not examined document traceability as a navigation and interaction problem in GMP-regulated environments, nor has it treated loopback behavior as an empirical indicator of document-structural friction. This study addresses both gaps.

3 Method

Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. GMP document environments are characterized by interdependent document networks — SOPs, CAPA records, analytical method protocols, validation artifacts, and specification documents — in which cross-references are textual rather than navigable, and the burden of reconstructing pathways falls on practitioners. The study examines two related questions: whether making linkage structure visible reduces traceability burden (Study 1), and whether behavioral navigation tracking can surface document-structural friction as an observable, measurable signal (Study 2).

Participants

Both studies were conducted with the same participant pool of $N = 20$, recruited from the same pharmaceutical quality team. All participants had GMP work experience with daily procedural document use; most had direct experience with deviation or OOS investigation workflows. Recruiting the same participants for both studies allowed for consistent baseline familiarity with the document environment, but may have introduced shared response tendencies across evaluations. This limitation is acknowledged in the scope section below.

Synthetic Document Corpora

Both studies used synthetic document corpora constructed in Python. Because actual facility documents are subject to non-disclosure obligations, corpora were designed to reflect the structural and terminological characteristics of real GMP documentation without reproducing proprietary content.

For Study 1, the corpus represented an interdependent network of SOPs, CAPA records, work instructions, validation artifacts, and specification documents, with explicit cross-references and lifecycle state assignments. For Study 2, the corpus consisted of seven synthetic GMP documents — an SOP, an analytical method protocol, a reference specification, a validation record, a training record, an OOS procedure, and a deviation form — structured to reflect a realistic OOS investigation workflow, with intentional cross-reference fragmentation to create naturalistic navigation demands.

Study 1: Traceability Visibility Prototype

Document Network Analysis

The Study 1 corpus was modeled as a directed graph using Python and NetworkX, with documents as nodes and cross-references as directed edges. The graph analysis identified three categories of linkage problem: missing links (references implied by document content but absent from the metadata structure), outdated references (links pointing to superseded or inactive documents), and lifecycle mismatches (documents whose version state conflicted

with the currency expectations of their referencing documents). The synthetic network yielded 31 missing CAPA links, 103 broken downstream links, and 21 lifecycle mismatch risks.

Prototype Development

Based on the network analysis, I developed an external traceability visibility prototype as a Streamlit web application. The prototype presents the document corpus as a navigable directed graph, allowing practitioners to trace upstream and downstream reference chains, identify broken or outdated links, and inspect lifecycle state mismatches. The system operates over the metadata structure; it does not modify, version, or interact with the synthetic source documents.

Evaluation

Participants completed three task-based traceability scenarios: locating the next downstream document in a reference chain, determining whether a given pathway was complete, and identifying where a reference had become outdated or broken. Post-task measures included task completion time, path completion accuracy, document discovery rate, ease of use, and adoption intent.

Study 2: Behavioral Navigation Tracking

Scenario Design

Participants were given a standardized OOS investigation scenario — an ELISA assay result exceeding specification — and asked to navigate across the seven synthetic documents to identify the relevant procedures and supporting records. No guidance or navigation hints were provided. The scenario was designed so that a complete investigation pathway required visiting multiple document types in a sequence that is non-obvious without prior familiarity with GMP investigation conventions.

Behavioral Tracking Prototype

I developed a navigation tracking prototype that records, for each participant session: document-to-document transitions with action type (forward navigation, cross-reference jump, or loopback); visit count and time spent per document; and the full navigation sequence as a directed path. At the end of each session, the prototype generates a directed network graph of the participant's navigation path and exports session data as a CSV file. The system tracks navigation behavior without modifying any source document.

A loopback was operationally defined as any return to a document that had already been visited within the same session, where the intervening path had visited at least one other document. This definition excludes immediate back-navigation (which may reflect interface behavior rather than structural reorientation) while capturing returns that indicate unresolved judgment demand.

Evaluation

Post-task measures included document list clarity, content readability, cross-reference clarity, navigation awareness, and real-world utility. Qualitative feedback probed the sources of navigation difficulty — specifically, whether participants attributed confusion to the prototype interface or to the structure of the documents themselves.

Scope and Limitations

Both studies are exploratory and pilot in scope. The use of the same participant pool across both studies limits the independence of findings and may have inflated usability ratings through familiarity effects. Synthetic corpora underrepresent the terminological variance and structural complexity of live GMP archives. The loopback operationalization used here is a first approximation; future work should examine whether different thresholds or definitions produce meaningfully different results. Neither study establishes effects on downstream outcomes — investigation quality, time-to-closure, or regulatory compliance.

4 Findings

Study 1: Traceability Visibility

Cross-Reference Visibility as Handover Infrastructure

The most consequential participant response to the traceability visibility prototype concerned not routine document navigation but personnel handover — a condition that occurs frequently in pharmaceutical quality operations and that current document systems handle poorly. When a quality practitioner changes roles or leaves a team, the change control records, deviation reports, and CAPA documents they managed do not transfer with any navigable account of their relationships. The incoming practitioner must reconstruct the full history of interconnected quality events manually, navigating a quality management system in which each document type may have a different owner and a different location, with relationships recorded only in document text rather than as traversable links.

Participants described this reconstruction work as one of the most time-consuming and error-prone aspects of their roles. The traceability visibility prototype addressed it directly: by making cross-reference relationships visible as a navigable graph, it allowed the full chain of related quality events to be seen at once. Participants described this as transformative for handover situations, and several anticipated that it would reduce the time required to become effective in a new role.

A practical design question also emerged: what happens when two or more quality events are cross-linked — when a single deviation is connected to multiple CAPAs, or when a change control record references both an ongoing deviation and a completed CAPA from a prior cycle? Participants noted that the current prototype handles single-chain navigation well but would need to support multi-event tagging for more complex relationship structures. This points to a design extension rather than a design failure.

Task Performance

In task-based evaluation, the traceability visibility prototype was associated with a 30–60% reduction in path completion time, higher rates of complete path reconstruction, and

improved detection of broken and outdated references. Participants rated the system as easy to use and consistently indicated willingness to use it in similar work contexts.

Study 2: Behavioral Navigation Tracking

Loopback as a Structural Signal

Loopback behavior was observed across all twenty participant sessions. The most common pattern was: Analyst Training Record → ELISA Test Method SOP → Method Validation Report → ELISA Test Method SOP → OOS Investigation Procedure. This pattern reflects a recognizable investigative logic: a practitioner investigating an OOS result checks analyst training as a potential root cause, moves to the method SOP to verify procedure, checks validation to confirm method acceptability, then returns to the SOP to re-examine a specific step before proceeding to the formal OOS procedure. The return to the ELISA SOP — the loopback — is not evidence of confusion; it is evidence that the first visit did not fully resolve the judgment demand.

Interface usability ratings were consistently positive across all participants. Less experienced participants specifically identified SOP structure — ambiguous cross-references, non-obvious sequencing, overlapping scope across documents — as the source of navigational uncertainty. This dissociation between interface usability and navigation difficulty supports the central methodological claim: loopback behavior is a signal of document-structural friction, not interface friction.

Experience and Navigation Strategy

Participants with five or more years of GMP investigation experience showed markedly different navigation patterns from less experienced participants. Experienced practitioners typically opened a document related to analyst performance first — reflecting the practical heuristic that pipetting errors and reagent preparation are the most common sources of ELISA OOS results — and navigated with fewer transitions and fewer loopbacks. Less experienced participants were more likely to open the reference specification first, and exhibited more loopbacks as they attempted to reconstruct the investigative pathway from

structural cues rather than from internalized practice knowledge.

This experience asymmetry has direct implications for document system design. The navigation burden that experienced practitioners have reduced through years of practice remains fully present for less experienced ones — and is borne entirely by the practitioner, not redistributed or reduced by the document system.

The Seven-Document Structure

The scenario was constructed around seven documents because this reflects the approximate scope of a real GMP root cause analysis using the 4M+1E framework applied to an analytical OOS result. That a loopback to the ELISA SOP appeared across all sessions — even among experienced participants — suggests that the document system does not currently provide sufficient structural guidance to support linear navigation through even this minimal set.

Cross-Study Synthesis

Together, the two studies establish a structural-behavioral relationship. Study 1 shows that document linkage is incomplete and that making it visible reduces reconstruction burden. Study 2 shows that practitioners respond to incomplete linkage through navigation patterns — particularly loopbacks — that are consistent across sessions and reflect document structure rather than interface design or individual error. The two findings converge on a single claim: traceability failure does not remain at the level of documentation. It propagates into behavior, and that behavioral trace is empirically observable.

Both studies also converge on a finding about equity. The practitioners who bear the greatest navigation burden are those with the least accumulated implicit knowledge of the document system. The structural invisibility of document relationships is not experienced equally across a workforce. Making that structure visible is not only a usability improvement but a redistribution of epistemic access.

5 Discussion

Traceability as an Interaction Problem

The central contribution of this study is a reframing: traceability in regulated environments is not only a documentation property but a condition of interaction. When document relationships are only partially visible, the burden of reconstructing pathways is redistributed to practitioners, and that redistribution is not neutral. It falls disproportionately on those with the least accumulated implicit knowledge of the document system: new staff, incoming handover recipients, and investigators navigating document types they have not previously encountered. Making linkage structure visible does not merely improve efficiency; it changes who bears the cost of structural incompleteness.

This extends Harper's (1998) account of documents as organizational action. The present study shows what happens when those linkage conditions fail: the articulation work of bridging incomplete document networks becomes individually absorbed, invisible to quality systems, and unequally distributed across a workforce stratified by experience.

Loopback as Method

The methodological contribution of this study is the operationalization of loopback behavior as an empirical indicator of document-structural friction. Prior work on navigation behavior has treated loopbacks primarily as usability signals — evidence that an interface is confusing. The present study offers a different interpretation, grounded in information foraging theory (Pirolli & Card, 1999): a loopback is a rational response to weak information scent, not a navigational error. When a document does not provide sufficient cues to support forward navigation, returning to a prior document is the appropriate behavior.

This reframing has methodological implications beyond the present study. If loopbacks are structural signals rather than usability signals, they can be used as a diagnostic instrument for document system evaluation: by tracking where loopbacks occur in a navigation session, designers can identify the specific documents or cross-reference junctions producing the most friction — without deploying instruments inside validated systems.

Articulation Work and the Invisibility of Navigation Burden

The handover finding from Study 1 illustrates the mechanism that Schmidt & Bannon (1992) describe as articulation work. When a quality practitioner changes roles, the document relationships they have accumulated do not transfer with them. The incoming practitioner must reconstruct these relationships manually — real labor that is invisible to any quality metric, unattributed to the structural conditions that produce it, and unrecognized as a cost by the organizations that depend on it.

The traceability visibility prototype makes this invisible labor visible by externalizing the relationship structure that practitioners currently hold in implicit knowledge. It does not eliminate the need for expertise; it reduces the extent to which expertise is required for basic navigational work that the document system should support structurally.

Epistemic Access in Regulated Work

The convergence of findings across both studies points toward a broader framing: document system design in regulated environments is, among other things, a problem of epistemic access — of who can reconstruct defensible pathways, under what conditions, and at what cost. The experience asymmetry documented in Study 2 is not a training gap. It is a design gap. Experienced practitioners have internalized a map of the document system that the system itself does not provide. Making linkage structure visible is a way of externalizing that map — of building into the document system the navigational support that currently exists only in the heads of experienced practitioners.

Design Implications

Two design principles emerge from the combined findings. First, traceability visibility should be treated as a first-class design concern in GMP document systems, not as a secondary maintenance task. The graph-based visibility approach demonstrated here is technically straightforward and does not require modifying validated source files. The primary barrier to adoption is conceptual: document systems in regulated environments

are typically designed for compliance rather than for navigation.

Second, behavioral navigation tracking should be adopted as an evaluation method for document system design. Loopback frequency, path length, and the location of reorientation events are tractable metrics that can be collected through external prototypes and used to identify structural friction before it is absorbed as individual burden.

Limitations and Future Work

The most important limitation of this study is the use of synthetic corpora. Both studies used document sets generated to reflect real GMP structures but lacking the full terminological variance, document volume, and cross-reference complexity of live archives. Future work should deploy both prototypes against live document systems in production quality environments. The handover use case identified in Study 1 is a particularly promising deployment scenario: personnel transitions are frequent, the navigation burden they impose is well recognized by practitioners, and a traceability visibility layer addresses a problem that has no current structural solution. Future work should also examine whether loopback reduction, once achieved through visibility improvements, persists over time or whether practitioners develop new navigational strategies that produce different friction patterns.

6 Conclusion

This paper introduced document traceability as a condition of interaction in regulated environments — not only a documentation property, but a structural determinant of who can navigate defensible pathways, at what cost, and under what conditions. When cross-reference relationships are invisible, the labor of reconstructing them falls to individual practitioners, distributed unequally across a workforce stratified by experience. Making linkage structure visible does not merely reduce search time; it redistributes epistemic access.

Two prototype-based studies establish this claim empirically. Study 1 demonstrated

that a graph-based traceability visibility layer — operating externally over a synthetic GMP document corpus without modifying validated source files — reduced path reconstruction time by 30–60% and was anticipated as transformative for personnel handover situations. Study 2 demonstrated that loopback behavior — observed across all twenty participant sessions in a standardized OOS investigation scenario — is attributable to document structure rather than interface design, and that less experienced practitioners bear a disproportionate share of the navigational burden that structural incompleteness produces.

Three contributions follow. First, document traceability is introduced as a first-class HCI design problem in regulated environments, where the no-modification constraint creates a design boundary with no precedent in existing navigation research. Second, loopback tracking is proposed as an empirical method for surfacing document-structural friction without deployment inside validated systems — a diagnostic instrument applicable beyond this study. Third, the theoretical framing — traceability as a condition of epistemic access — connects HCI concerns about document navigation to broader organizational questions about how accountability is structured in regulated sociotechnical work.

The most important direction for future work is deployment with live documents in production quality systems, where the full implications of traceability visibility — for navigation efficiency, for the distribution of invisible labor, and for the equity of epistemic access across a workforce — can be assessed in real use.

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